

DATA STRUCTURE

AGENDA

- ☐ What Are Data Structures?
- ☐ Linear vs Non-Linear Structures
- ☐ Arrays
- ☐ Linked Lists
- ☐ Stack (LIFO)
- ☐ Queue (FIFO)
- ☐ Hash Tables
- ☐ Trees & Binary Search Trees
- ☐ Graphs
- ☐ Real-World Applications
- ☐ Introduction to Algorithms

❏ WHAT ARE DATA STRUCTURES?

□ **Data Structures:**

Data structures are specialized formats used to organize, manage, and store data in a way that enables efficient access and modification. They form the foundation for writing efficient algorithms and solving complex problems. There are two main categories:

- **Linear Data Structures**, where data elements are arranged in a sequential order.
- **Non-Linear Data Structures**, where elements are organized hierarchically or in a network.

□ LINEAR VS NON-LINEAR

❑ Classification of Data Structures:

1. Linear Data Structures:

Elements are stored one after another.

They follow a specific order: first, next, last.

Examples: **Array, Linked List, Stack, Queue, Hash Table.**

2. Non-Linear Data Structures:

Elements are stored in a hierarchical or networked fashion.

Not stored sequentially; parent-child or interconnected relationships exist.

Examples: **Tree, Graph, Heap.**

□ Array:

An **array** is a fixed-size collection of elements, all of the same type, stored in contiguous memory locations.

Each element can be accessed directly by its index.

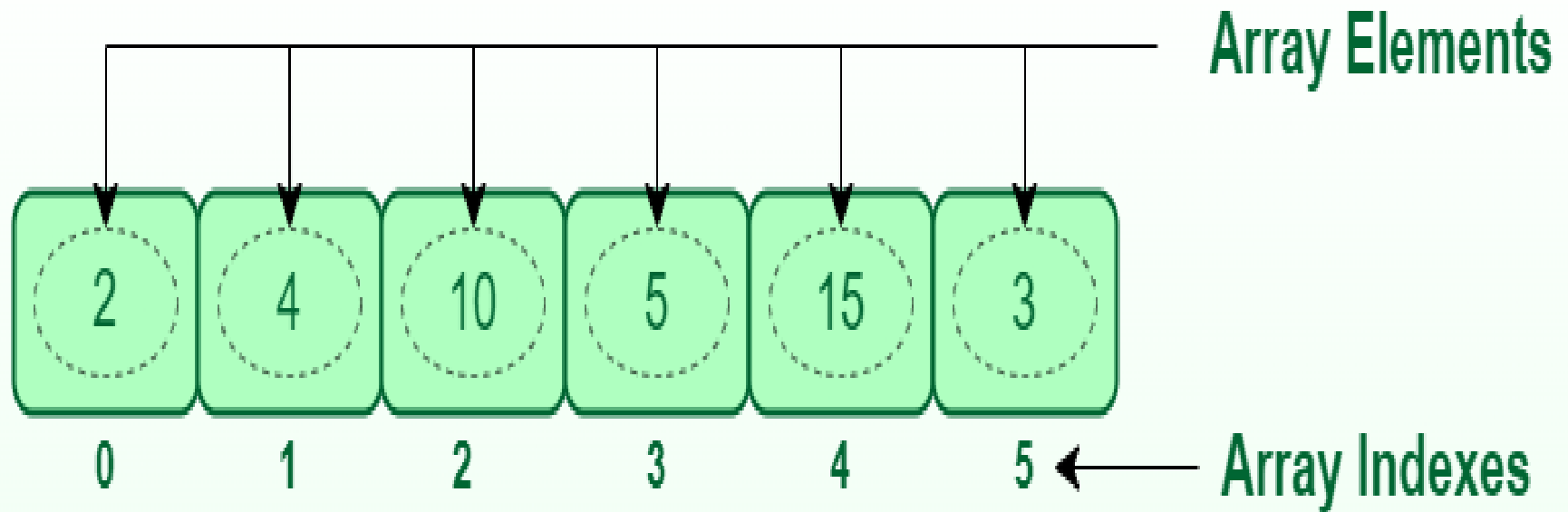
Key Properties:

- Fast element access using indexes.
- Static memory allocation (size is fixed at creation).

Uses:

- Storing data when the size is known.
- Lookup tables, image processing, matrices.

Array



□ Stack (LIFO)

A **stack** is a linear data structure that follows the Last-In, First-Out (LIFO) principle. The most recent item added is the first to be removed.

Operations:

- Push: Add item
- Pop: Remove item
- Peek: View top item

Uses:

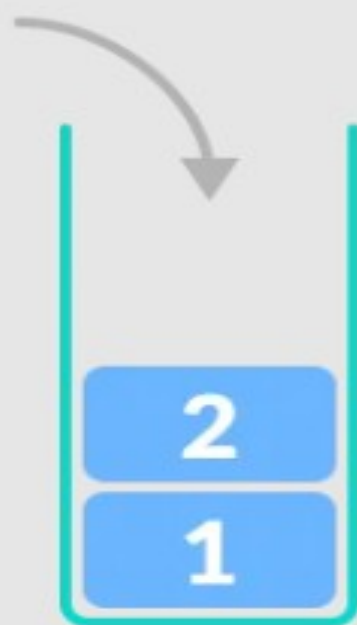
- Undo mechanisms in text editors.
- Backtracking (e.g., maze solvers).
- Function call management in recursion.



**empty
stack**



push



push



push



pop

❑ Queue (FIFO)

A **queue** is a linear structure that follows the First-In, First-Out (FIFO) principle. The first element added is the first to be removed.

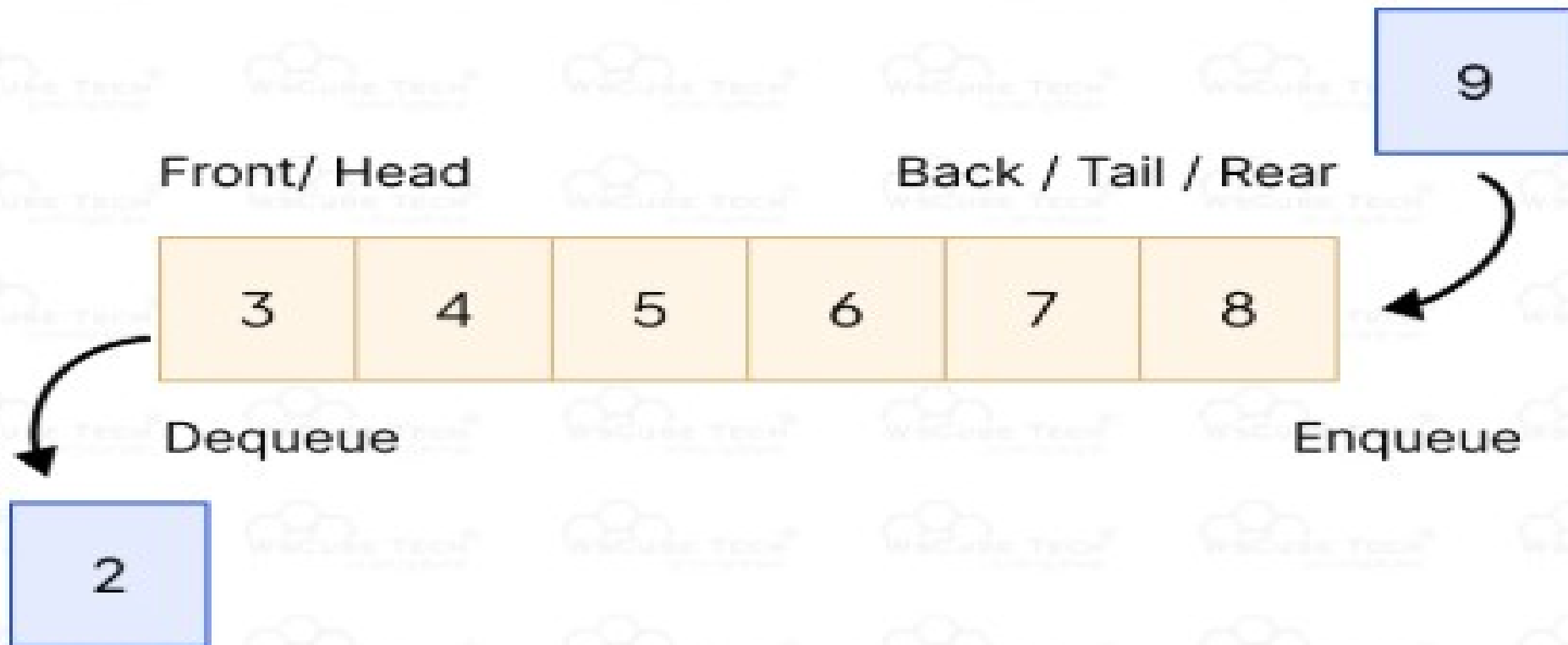
Operations:

- Enqueue: Add item to the back.
- Dequeue: Remove item from the front.

Uses:

- Task scheduling (CPU, printing).
- Data streaming.
- Buffers in I/O systems.

Queue Data Structure



❑ **Linked List:**

A **linked list** is a dynamic data structure where each element (called a node) contains data and a reference (link) to the next node.

Key Properties:

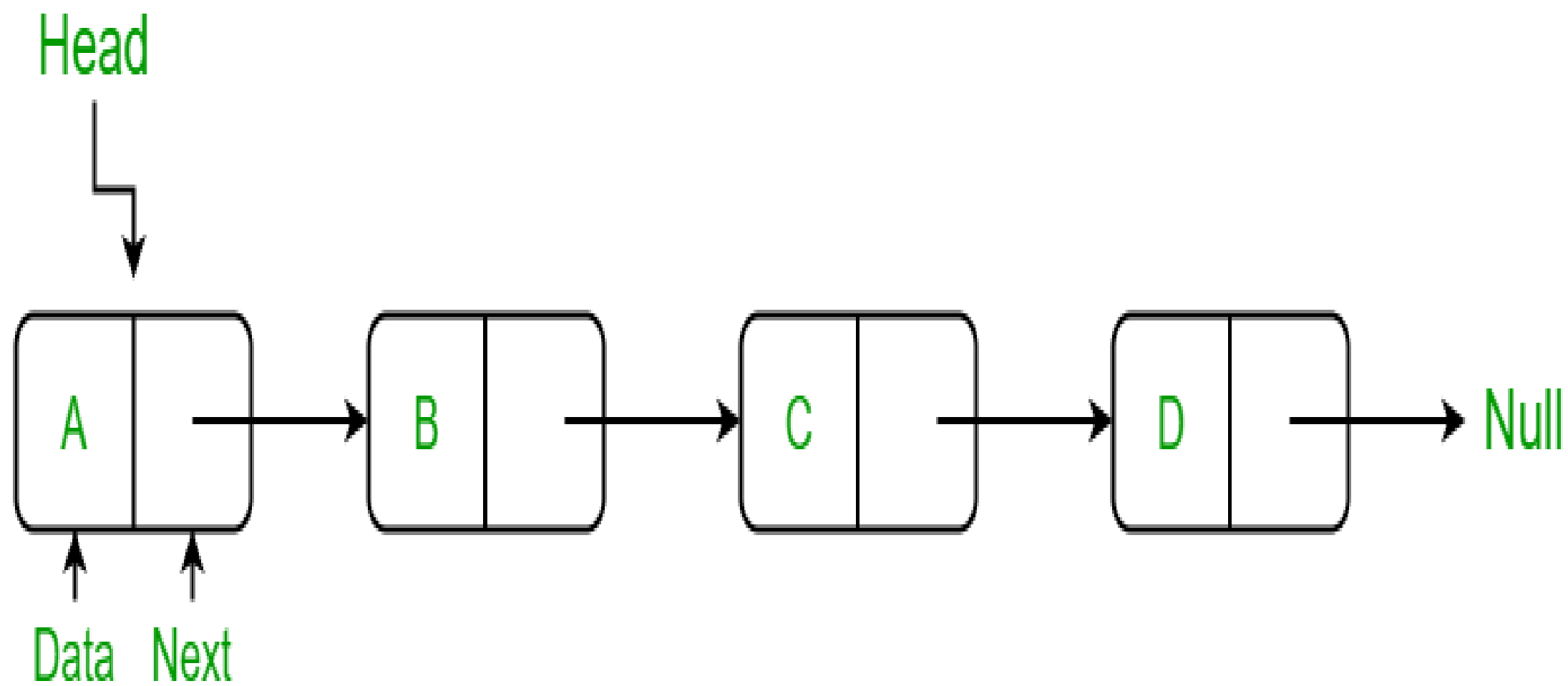
- Dynamic size
- Efficient insertions and deletions ($O(1)$ for front)
- Slower access ($O(n)$ for search)

Types:

- Singly Linked List
- Doubly Linked List
- Circular Linked List

Uses:

- Implementing stacks/queues.
- Navigating music playlists or image galleries.



□ Hash Table:

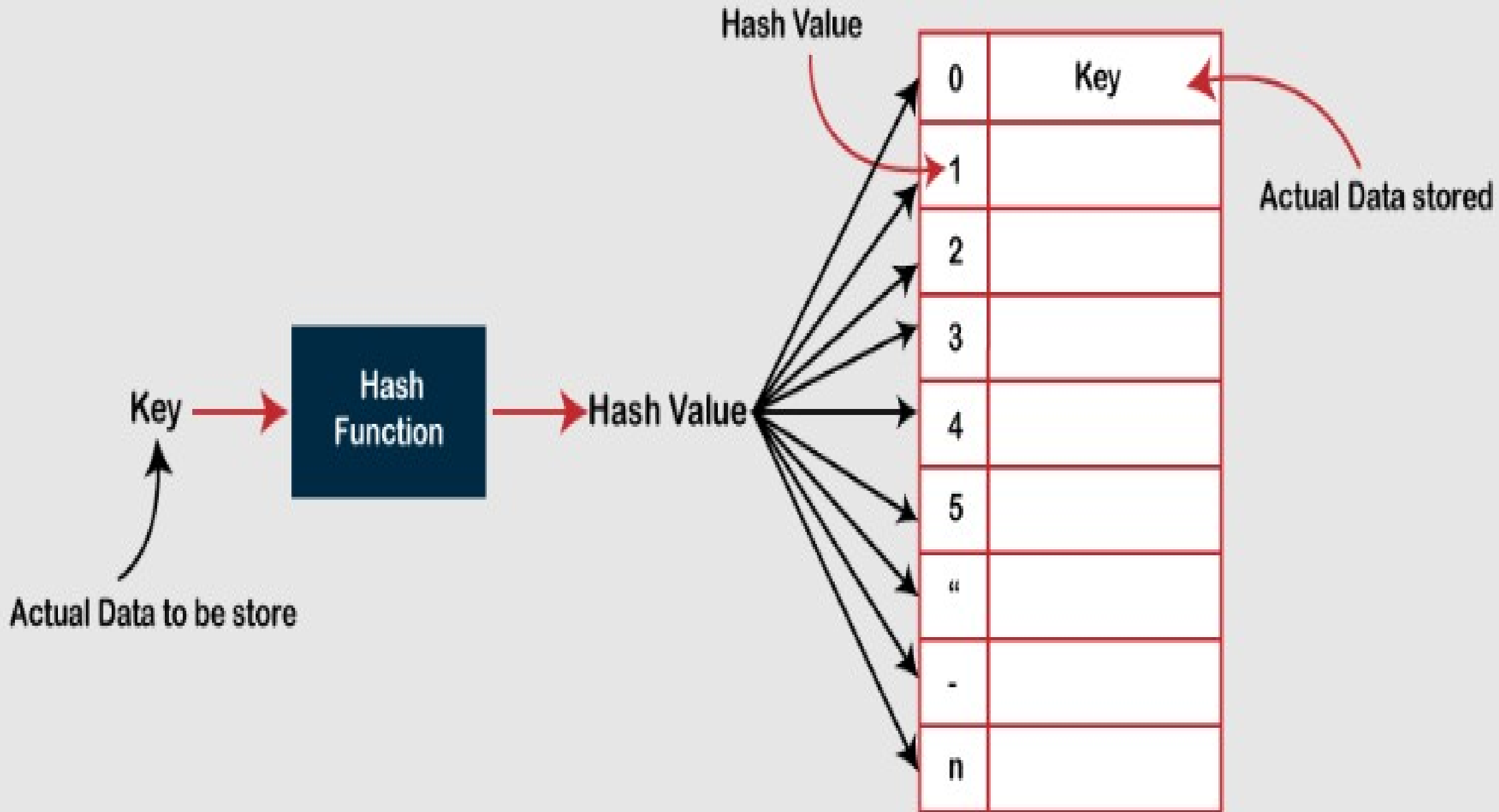
A **hash table** stores data in key-value pairs. A hash function converts the key into an index to store the value.

Key Properties:

- Very fast search, insert, delete ($O(1)$ average).
- Collisions can occur and are handled using chaining or open addressing.

Uses:

- Fast lookups (user data, cache, symbol tables).
- Implementing sets and maps in programming



□ Graph:

A **graph** is a collection of **nodes (vertices)** connected by **edges**. Graphs may be:

- Directed or Undirected
- Weighted or Unweighted

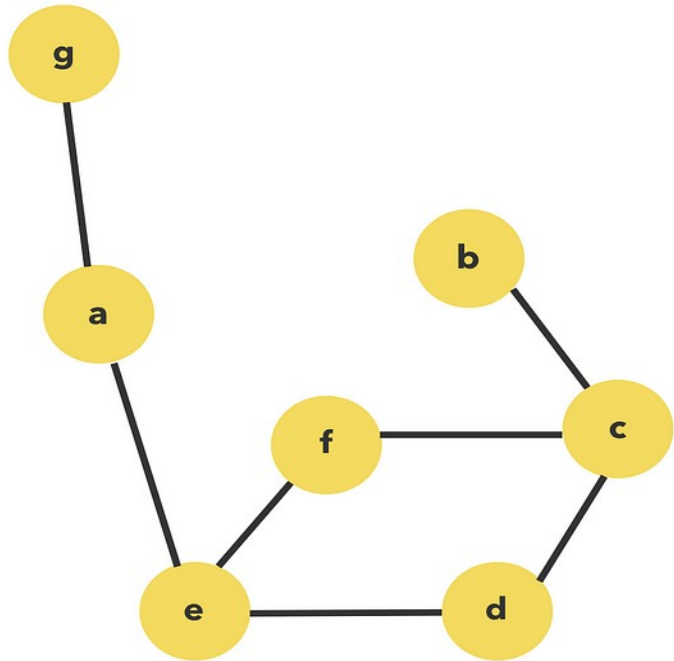
Real-Life Example:

A **map** of cities and roads, or social networks like Facebook (users connected as friends).

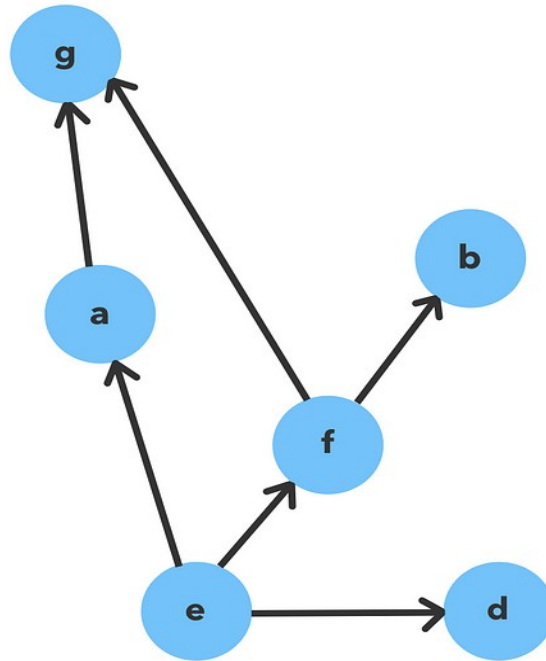
Uses:

- Navigation systems (Google Maps)
- Social networks (suggested friends)
- Web page ranking (Google's PageRank)

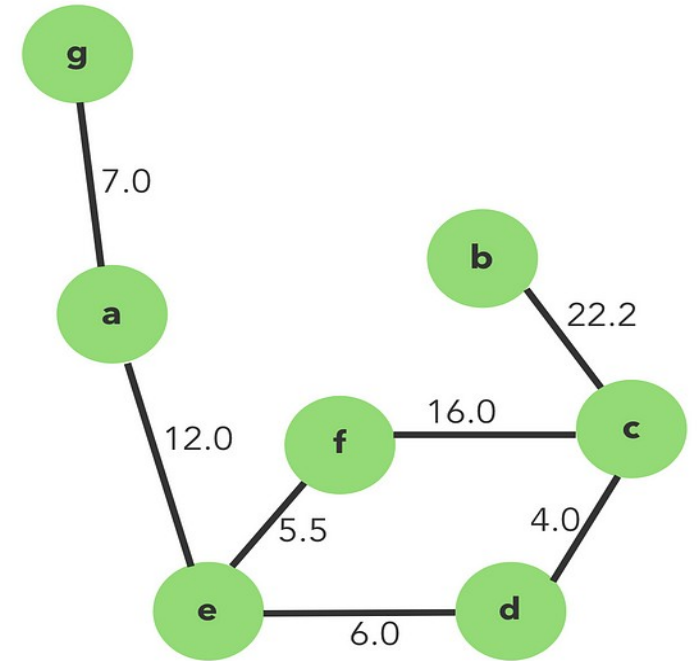
Undirected



Directed



Weighted



□ Tree:

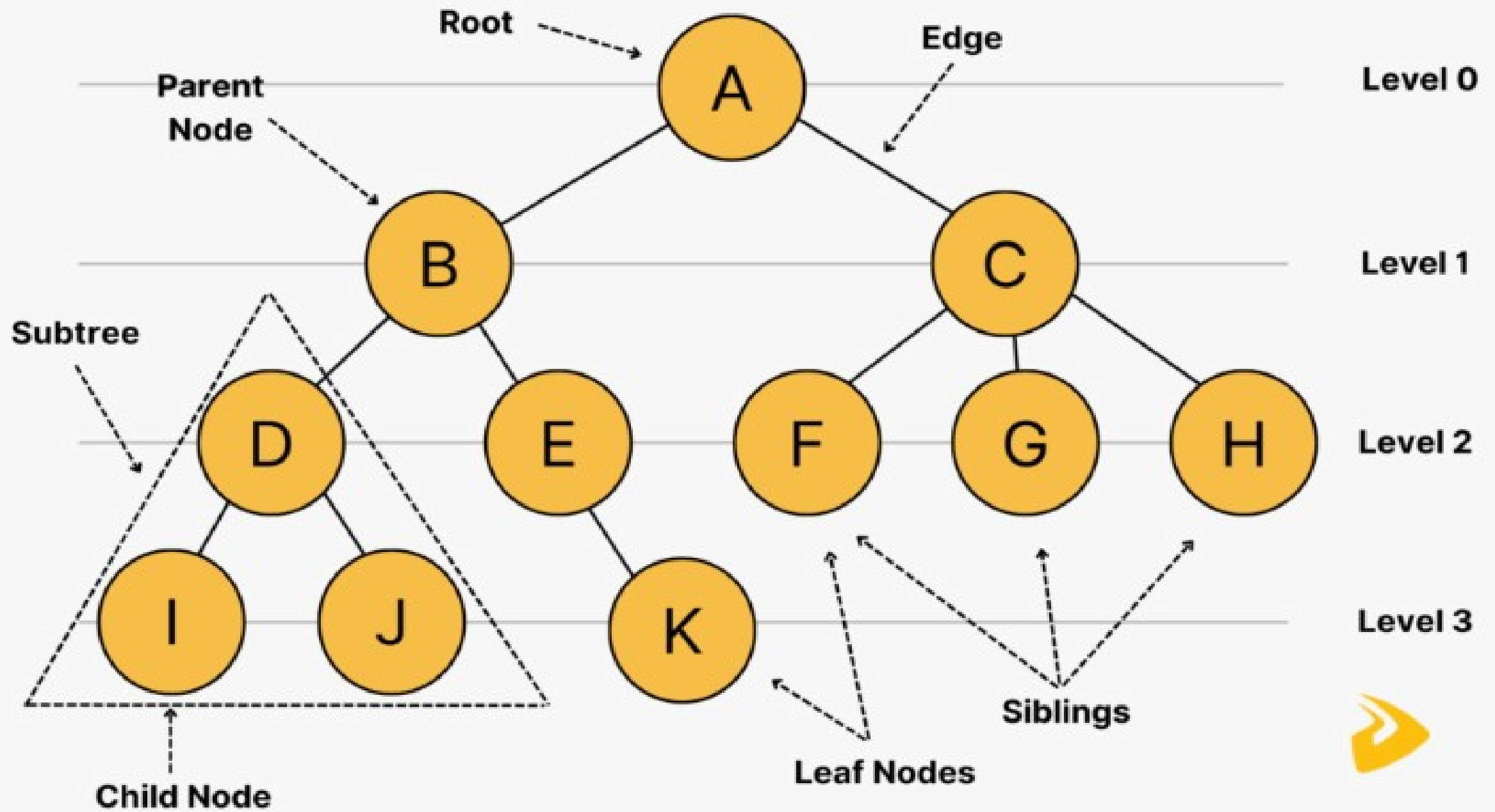
- A **tree** is a hierarchical data structure made up of nodes. One node is designated as the **root**, and every other node is connected through **parent-child** relationships.

Real-Life Example:

A **family tree** or a **folder system** on a computer.

Uses:

- File organization systems.
- Routing algorithms.
- Database indexing.



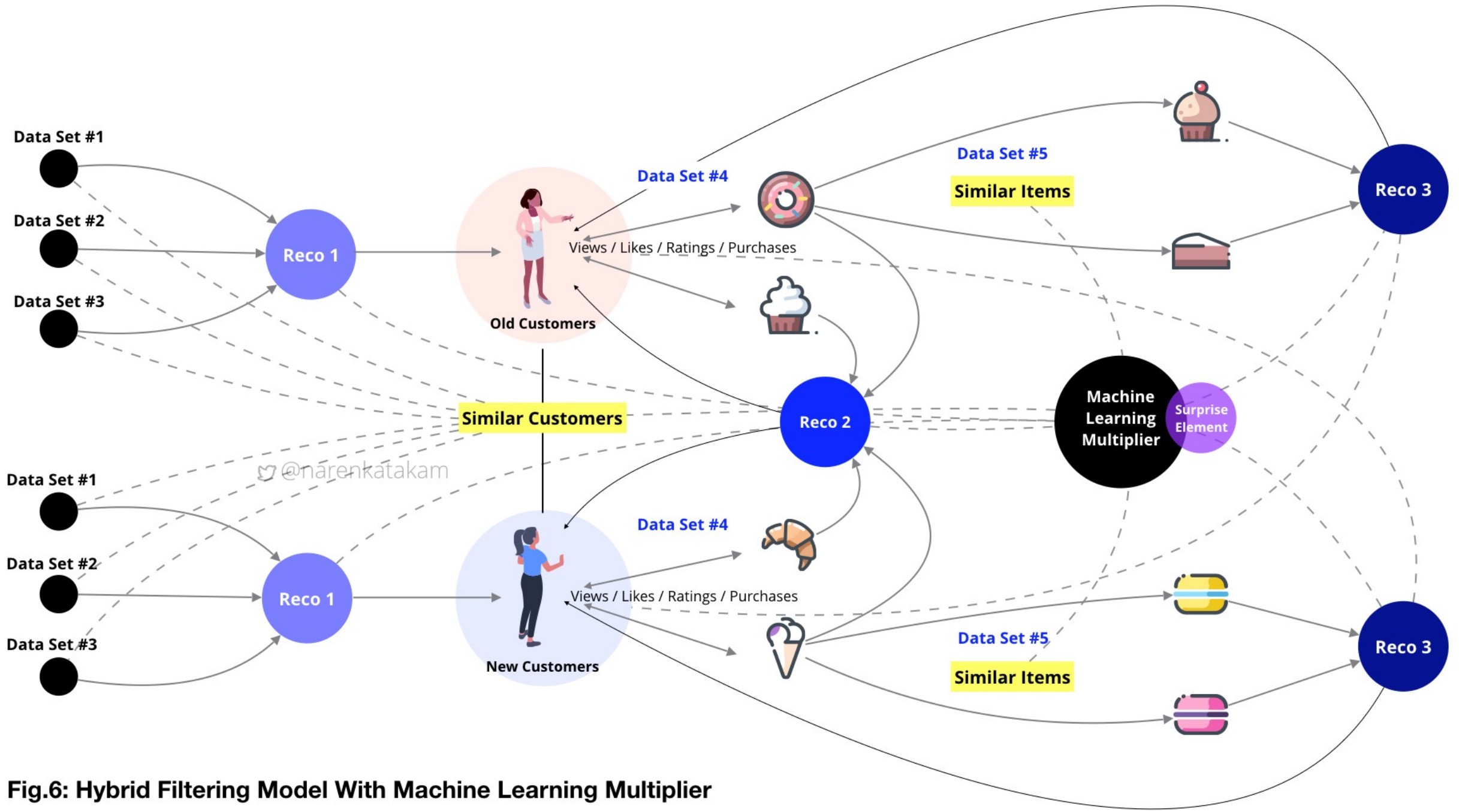
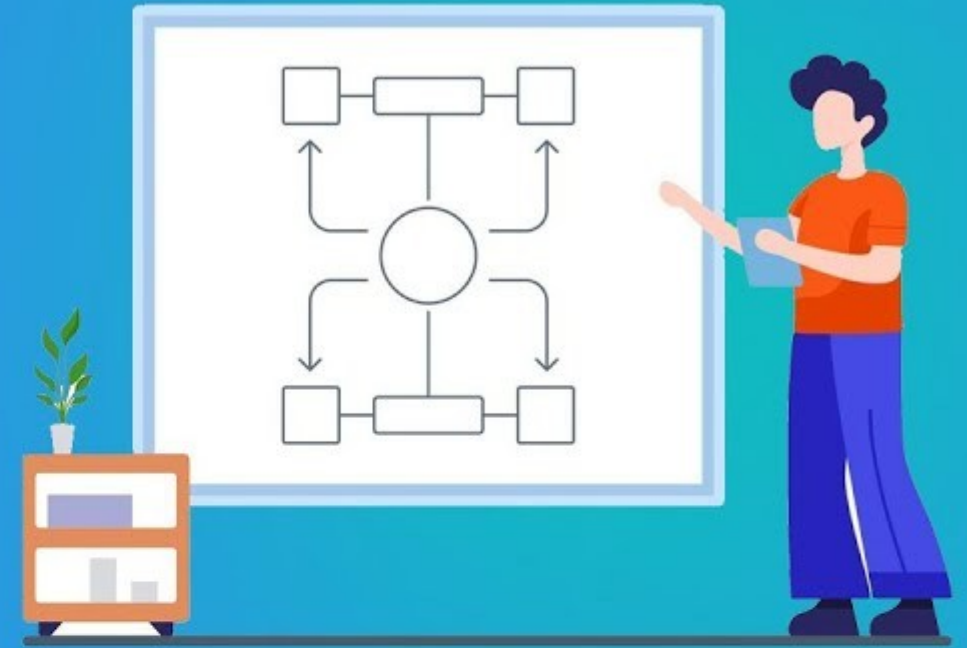


Fig.6: Hybrid Filtering Model With Machine Learning Multiplier



WHAT IS AN ALGORITHM?

❏ QUESTIONS ?